

SYSC 5108 Exam Study

Assignment 4, Question 6

Jesse Levine

Question

Given the two-dimensional dataset

$$\begin{array}{c|ccccc} x_1 & 2 & 0 & -3 & 3 & -2 \\ x_2 & -2 & 4 & -1 & -3 & 2 \end{array}$$

find:

- (a) the one-dimensional PCA,
- (b) the encoded points,
- (c) the reconstructed points,
- (d) the percentage of variance explained.

Course and Textbook Cross-References

- **Chapter 7 slides, PCA:** PCA finds low-dimensional components that preserve as much variance as possible.
- **Chapter 7 slides, Linear PCA and Interpretation:** the principal directions come from the covariance structure of the data.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, PCA derivation:** the encoding is a linear projection onto principal directions, and the reconstruction is obtained from the projected code.

Step 1: Write the Data Points

The five samples are

$$x^{(1)} = (2, -2), \quad x^{(2)} = (0, 4), \quad x^{(3)} = (-3, -1), \quad x^{(4)} = (3, -3), \quad x^{(5)} = (-2, 2).$$

Step 2: Compute the Sample Mean

The mean is

$$\mu = \frac{1}{5} \sum_{i=1}^5 x^{(i)} = \frac{1}{5} \begin{bmatrix} 2 + 0 - 3 + 3 - 2 \\ -2 + 4 - 1 - 3 + 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

So the data are already centered.

Step 3: Form the Data Matrix

Using columns as data points,

$$X = \begin{bmatrix} 2 & 0 & -3 & 3 & -2 \\ -2 & 4 & -1 & -3 & 2 \end{bmatrix}.$$

Step 4: Compute the Covariance Matrix

Because the mean is zero, the sample covariance matrix used in the submitted solution is

$$\Sigma = \frac{1}{5}XX^\top.$$

First compute $XX^\top$:

$$XX^\top = \begin{bmatrix} 2 & 0 & -3 & 3 & -2 \\ -2 & 4 & -1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 0 & 4 \\ -3 & -1 \\ 3 & -3 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 26 & -14 \\ -14 & 34 \end{bmatrix}.$$

Therefore

$$\Sigma = \frac{1}{5} \begin{bmatrix} 26 & -14 \\ -14 & 34 \end{bmatrix} = \begin{bmatrix} 5.2 & -2.8 \\ -2.8 & 6.8 \end{bmatrix}.$$

Step 5: Find the Principal Eigenvalue and Eigenvector

The principal component is the eigenvector of Σ corresponding to the largest eigenvalue.

Solve

$$\begin{aligned} \det(\Sigma - \lambda I) &= 0 : \\ \det \begin{bmatrix} 5.2 - \lambda & -2.8 \\ -2.8 & 6.8 - \lambda \end{bmatrix} &= 0. \end{aligned}$$

So

$$\begin{aligned} (5.2 - \lambda)(6.8 - \lambda) - (-2.8)^2 &= 0, \\ \lambda^2 - 12\lambda + 27.52 &= 0. \end{aligned}$$

The two eigenvalues are

$$\lambda_1 \approx 8.9120, \quad \lambda_2 \approx 3.0880.$$

The largest eigenvalue is λ_1 , so the one-dimensional PCA direction is its unit eigenvector:

$$u_1 = \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix}.$$

Answer to Part (a): One-Dimensional PCA

$$u_1 = \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix}$$

Note: the sign can be flipped and still represent the same principal direction.

Step 6: Encode the Data Points

For one-dimensional PCA, each encoded coordinate is the projection

$$z_i = u_1^\top x^{(i)}.$$

Compute each one:

$$z_1 = [0.6022 \quad -0.7983] \begin{bmatrix} 2 \\ -2 \end{bmatrix} = 2.8011,$$

$$z_2 = [0.6022 \quad -0.7983] \begin{bmatrix} 0 \\ 4 \end{bmatrix} = -3.1934,$$

$$z_3 = [0.6022 \quad -0.7983] \begin{bmatrix} -3 \\ -1 \end{bmatrix} = -1.0082,$$

$$z_4 = [0.6022 \quad -0.7983] \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 4.2016,$$

$$z_5 = [0.6022 \quad -0.7983] \begin{bmatrix} -2 \\ 2 \end{bmatrix} = -2.8011.$$

Answer to Part (b): Encoded Points

$$z = [2.8011 \quad -3.1934 \quad -1.0082 \quad 4.2016 \quad -2.8011]$$

Step 7: Reconstruct the Data

The PCA reconstruction formula is

$$\hat{x}^{(i)} = \mu + u_1 z_i.$$

Since $\mu = 0$,

$$\hat{x}^{(i)} = u_1 z_i.$$

Thus

$$\hat{x}^{(1)} = 2.8011 \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix} \approx \begin{bmatrix} 1.6868 \\ -2.2362 \end{bmatrix},$$

$$\hat{x}^{(2)} = -3.1934 \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix} \approx \begin{bmatrix} -1.9230 \\ 2.5494 \end{bmatrix},$$

$$\hat{x}^{(3)} = -1.0082 \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix} \approx \begin{bmatrix} -0.6072 \\ 0.8049 \end{bmatrix},$$

$$\hat{x}^{(4)} = 4.2016 \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix} \approx \begin{bmatrix} 2.5302 \\ -3.3544 \end{bmatrix},$$

$$\hat{x}^{(5)} = -2.8011 \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix} \approx \begin{bmatrix} -1.6868 \\ 2.2362 \end{bmatrix}.$$

Answer to Part (c): Reconstructed Points

$$\hat{x}^{(1)} = (1.6868, -2.2362), \quad \hat{x}^{(2)} = (-1.9230, 2.5494), \quad \hat{x}^{(3)} = (-0.6072, 0.8049),$$

$$\hat{x}^{(4)} = (2.5302, -3.3544), \quad \hat{x}^{(5)} = (-1.6868, 2.2362).$$

Step 8: Percentage of Variance Explained

For PCA, the variance explained by the first component is

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Substitute the eigenvalues:

$$\frac{8.9120}{8.9120 + 3.0880} = \frac{8.9120}{12} \approx 0.7427.$$

So the explained variance percentage is

$$0.7427 \times 100\% \approx 74.27\%.$$

Answer to Part (d): Variance Explained

$$\text{Explained variance ratio} \approx 0.7427$$

$$\text{Explained variance percentage} \approx 74.27\%$$

Final Answer Summary

- One-dimensional PCA direction:

$$u_1 = \begin{bmatrix} 0.6022 \\ -0.7983 \end{bmatrix}$$

- Encoded points:

$$z = [2.8011, -3.1934, -1.0082, 4.2016, -2.8011]$$

- Reconstructed points:

$$(1.6868, -2.2362), (-1.9230, 2.5494), (-0.6072, 0.8049), (2.5302, -3.3544), (-1.6868, 2.2362)$$

- Variance explained:

$$74.27\%$$

Exam Shortcut

For a small PCA question:

1. Center the data.
2. Compute the covariance matrix.
3. Take the eigenvector with the largest eigenvalue.
4. Encode with projection $z_i = u_1^\top x_i$.
5. Reconstruct with $\hat{x}_i = \mu + u_1 z_i$.
6. Compute explained variance as $\lambda_1 / \sum_j \lambda_j$.

Sources Used

- Assignment 4 PDF, Question 6.
- Local submitted Assignment 4 PDF checked for consistency of the numerical results.
- Chapter 7 slides on PCA, linear PCA, and the covariance interpretation.
- Goodfellow, Bengio, and Courville, *Deep Learning*, PCA derivation via projection and reconstruction.